

Performative Prediction in a Single-Asset Market

Self-fulfilment and signal erosion under forecast adoption

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0.07 → 0.19

what a live forecaster sees (rises)
realised-return R^2

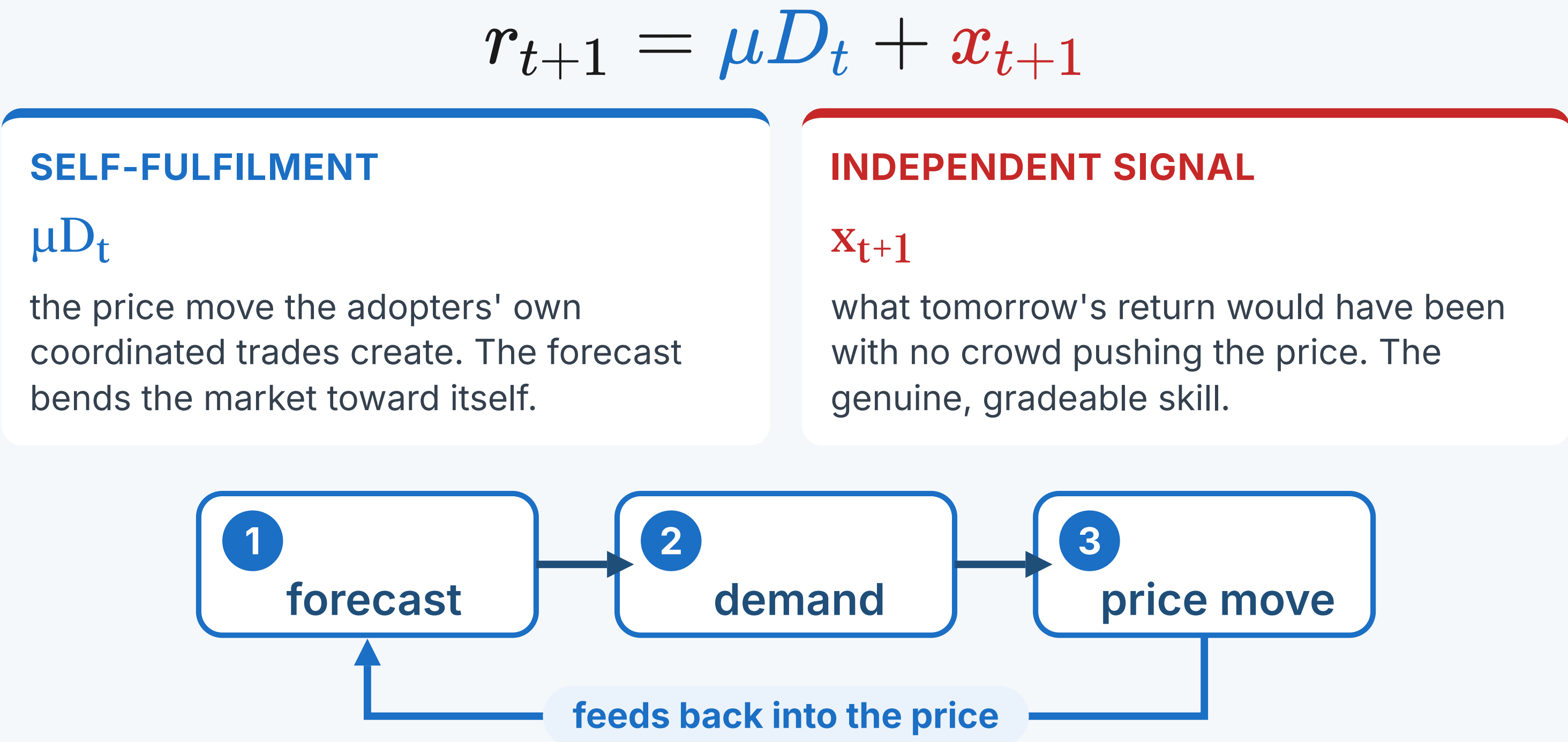
0.08 → 0.04

the genuinely independent signal (falls)
demand-adjusted R^2

0.28 → 0.45

the market's memory stiffens
effective persistence ϕ

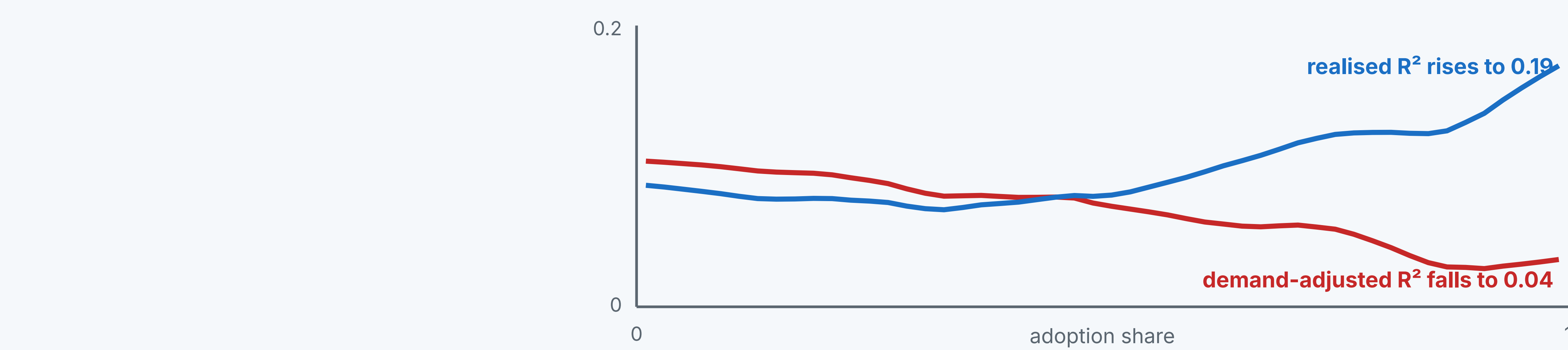
One forecast, two channels



One simulated market



As adoption rises, one forecast earns two verdicts



The question

If everyone trades on the same forecast, does it get better, or just bend the market into agreeing? Publish one forecast of tomorrow's return and traders begin acting on it. Their coordinated buying and selling pushes the very price the forecast will later be graded against.

The model

Strip a market to its essentials: traders who buy more when they expect a higher return and hold back when risk is high, and a market maker who moves the quote to absorb their net demand (Brock and Hommes 1998; Beja and Goldman 1980; Farmer and Joshi 2002). It is the smallest market in which a shared forecast can move the prices it is later judged against.

Three lines define the whole market: how returns form, how the price moves, and how each trader sizes a bet.

$$r_{t+1} = \varphi r_t + \mu D_t + \sigma \varepsilon_{t+1}$$

next return = momentum from last return, plus the crowd's price push, plus news

$$p_{t+1} = p_t + \mu D_t + \sigma \varepsilon_{t+1}$$

the market maker nudges the quote by that same crowd push plus news

$$z_{i,t} = \frac{\hat{r}_{i,t+1}}{a \sigma^2}$$

bet size = forecast return divided by risk aversion times variance

Every figure flows from one random seed: the whole simulator is reproducible.

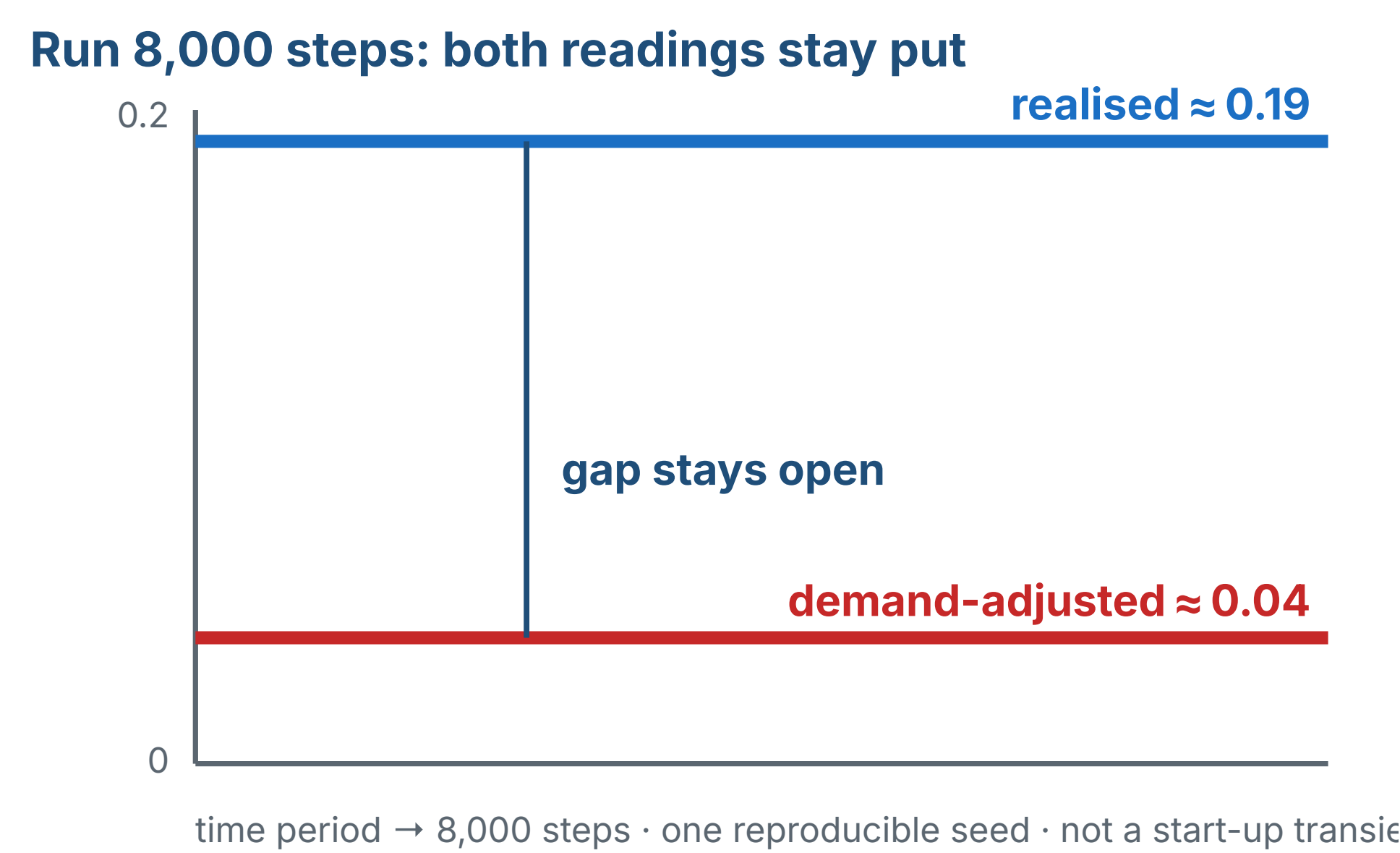
Intra-period timing

- observe returns through t , form forecasts
- adopters and non-adopters submit orders
- maker absorbs demand, moves the quote
- news shock and persistence term realise

Two evaluation targets

We score the same forecast two ways on the exact same price path. The realised-return score asks the deployment question: did the price actually move as predicted? The demand-adjusted score first strips out the adopters' own price impact (the μD_t term), asking the honest question: was the signal genuinely there before the crowd pushed? One is what a live forecaster measures, the other is the real underlying skill.

A steady state, not a transient



Realised rises by self-fulfilment; demand-adjusted erodes on the very same path.

The dual-channel result

Here is the twist. As more traders adopt the forecast, its realised-return score climbs from 0.07 to 0.19, looking sharper and sharper. But that rise is mostly self-fulfilment: the crowd's coordinated trades drag the price toward the prediction, so the forecast is partly grading its own homework. Strip that push out and the demand-adjusted score falls the other way, from 0.08 to 0.04, meaning the genuine, independent signal is actually eroding. The same trades stiffen the market's memory too (effective persistence rises from 0.28 to 0.45), feeding the self-fulfilling loop even harder.

Conclusion

Realised accuracy over-reports skill the moment a forecast has users who move the market. The gap between the two readings is not noise: it is the slice of measured accuracy manufactured by the forecast's own deployment. Report both readings, or mistake self-fulfilment for foresight.

Soros 1987 · Perdomo et al. 2020 · Brock & Hommes 1998 · Beja & Goldman 1980 · Farmer & Joshi 2002 · Diebold & Mariano 1995